Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Advanced multi-scale modeling and characterization for a predictable and experimentally validated assessment of plasticity and fracture in multilayered metallic interconnectors during thermal cycles*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		*Mechanical and Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	*Seminconductor Materials*	*Metal Materials*	*Materials Analysis/Evaluation Technologies*
Concept of Proposed Technology	*The project targets at (a) developing a theoretical and computational framework to predict the dynamic evolution of microstructures associated with the motion and interaction of defects, which could provide a unique opportunity to explore macroscopic deformation modes of multi-layer interconnectors. (b) developing novel experimental methodologies based on dynamic, multiscale and multiphysical analysis to investigate the metallization layer behavior under stress, which could provide a deeper and more comprehensive understanding of failure mechanisms in metallization layers. (c) developing machine learning approaches based on the input from experimental and theoretical results and enable more accurate predictions of device reliability and performance. (d) developing a prototype nano-scale stress measurement system for assessing the reliability of multi-layer interconnectors by combining experimental results and nano-mechanics models. The results of the project can be potentially used by the engineers from chip manufacturers to identify potential failures,*			

	to understand performance degradation, to predict the lifespan of microchips, and to optimize their design to increase the lifetime. The developed prototype system can be potentially used to assess the reliability of semiconductor chips, e.g., interfacial adhesion in multi-layered semiconductor packaging structures.
Rationale for Joint R&D Project	*Advanced chiplet technology demands greater heterogeneity in materials, interfaces, and length scales, while product cycles are increasingly compressed—making accurate, physics-based reliability prediction even more critical. Globally, leading companies such as Intel, TSMC, and NVIDIA are advancing wafter fabrication, 3D heterogeneous integration and chiplet-based architectures to address performance and power scaling. There is growing demand for quantitative prediction of canonical failure modes in thin-film/layered structures under thermal cycling—cracking, interfacial debonding, ratcheting, and voiding. International standardization activities, including IEEE initiatives on reliability modeling and JEDEC standards for packaging, are also reinforcing these trends. On the other hand, cutting-edge wafer fabrication and advanced packaging development are active in South Korea led by Samsung electronics and SK hynix, but practical evaluations remain biased toward continuum-level simplified models and standard accelerated reliability testing; consequently, precise multi-scale analysis, in-situ monitoring with temporal resolution, simultaneous multiphysical fields analysis, etc. are still lacking. The proposed technology provides an innovative physics-based analysis framework that fills this gap. International cooperation is essential to advance multiscale modeling and reliability assessment in advanced packaging and multilayer interconnects. Collaborative efforts such as building a joint simulation infrastructure with international perspectives globally can enable the development of large-scale, data-driven multiscale simulation frameworks. The required capabilities needed in this project are beyond any single laboratory, advanced metrology tools and diverse modeling experties are distributed gloabally. Cross-border cooperation enables shared access to scarce infrastructures, creation of common datasets and benchmarks and best usage of the multi-scale modeling and characterization. In addition, experiment–model integration through cross-lab validation will ensure that computational predictions are consistently verified under real reliability testing environments. The development of fracture mechanics-based predictive and experimental valided modes, and integration of such models with precise measurement technology will bring the reliability evaluation of metalic connectors in microchips at another level. Such cooperation will accelerate both technological innovation and industrial adoption worldwide.*
Key Words	*1. Multiscale-Modeling and Characterization* *2. Advanced Package* *3. Interconnect-Reliability* *4. Dislocation Dynamics* *5. Thin Film Plasticity*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	*To achieve predictable technique for deformation and failure mechanisms in multilayered integrated circuits, enabled by integrated advanced multiscale modeling and experiments. To use the obtained experimental results and nano-mechanics models for developing and optimizing a nano-scale stress measurement system.*			
R&D Target Specifications	*-Comprehensive investigation on dislocation interactions with grain boundaries (GBs) and passivation layers (PLs) under various conditions using proper models.* *-Well understood effect of microscopic defect interactions with GBs and PLs on macroscopic deformation modes via multiscale computational models.* *-Modeling of void nucleation/growth, failure, and crack propagation using multiple damage criteria in multiscale simulations.* *-Develop a unified multiscale computation model with high fidelity for optimization of multilayer interconnectors.* *-Multi-scale experimental in-situ testing in microscopy to understand mechanics and materials behavior of multilayer interconnectors under stress.* *-Advanced nanoanalytics to understand the compositional fluctuations and microstructures associated with defects.* *-Established experimental workflows for assessement involving in-situ monitoring and multiphysical analysis.* *-A experimentally validated predictiable model to enable more accurate predictions, optimized design of interconnectors with prediction of various failure modes.* *- A nano-scale stress measurement equipment capable of reliability evaluation for multi-layer structures of semiconductor microchips, optimized by combining experimental results and nanomechanics models.*			
R&D Plan	*WP1: Investigate dislocation interaction with internal grain boundaries and external passivation layers via atomistic modeling. The modes of interaction of dislocation with various GBs will be characterized. (Month 1 to 12)* *WP2: Mesoscale discrete dislocation dynamics (DDD) modeling of single crystalline thin films. In addition, a coupled DDD–FEM framework will be developed to incorporate damage criteria for capturing void nucleation/growth and related failure processes. (Month 7 to 24)* *WP3: Atomistically-informed continuum crystal plasticity modeling for polycrystalline thin films. A computationally efficient crystal plasticity model based on the optimization method will be used. (Month 19 to 36)* *WP4: Understand the nanomechanics and failure modes in the metallization layers using multiscale in-situ microscopy. The mechanisms influenced by multiple factors will be fully investigated systematically. (Month 7 to 30)* *WP5: Correlate the failure modes with compositional and microstructural information through advanced nanoanalytics from macro to atomic level. The obtained detailed information in this task will be correlated with findings from dynamic study, therefore help to clarify the failure modes. (Month 1 to 24)* *WP6: Optimization of experimental workflows for fracture failure assessement involving in-situ monitoring and multiphysical analysis. With the support of machine learning, automation of this determination of relevant techniques and optimized workflows is expected. (Month 13 to 30)* *WP7: Creation of machine learning enhanced, experimental validated predictable mode. (Month 19 to 36)* *WP8: Development of mechanical property measurement system and system testing. (Month 1 to 24)*			

	WP9: Solution development and optimization supported by the experimental results and nano-mechanics models. (Month 25 to 36) *WP10: Project management. (Month 1 to 36)*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The research institute will focus on experimental support to achieve a predictable understanding of deformation and failure in multilayer interconnectors. It will develop novel experimental methodologies based on dynamic, multiscale and multiphysical analysis to investigate the metallization layer behavior under stress, and establish a clear cause–effect linkage across length scales so that microstructural constraints and interface-dominated architectures can be evaluated in a consistent, physics-informed manner under representative loading conditions.* *-Multiscale in-situ testing in microscopes with external stress, understand mechanics and materials behavior.* *-Investigate the compositional fluctuations and micrstructural changes.* *-Connect experiments and modeling in an iterative loop, and validate the predictions made by machine learning enhanced model.* *-Use experimental results to tune and improve the models and keep the approach consistent from small-scale calculations to larger-scale simulations.* *-Co-deliver an experimentally validated predictive capability and actionable design guidance for optimizing multilayer interconnectors against multiple failure modes.* *-Support solution development of nano-scale stress measurement equipment using experimental methodology and results.*
Expected R&R of Korean SME Partner	*To evaluate the reliability of multi-layered structures utilized in hybrid bonding and redistribution layers (RDL) of semiconductor packages, the development of fracture mechanics-based modeling combined with precise micro-measurement technology is essential. Therefore, the major role of Korean SME is to develop a nano-scale stress measurement equipment integrated with facture mechanics models capable of characterizing thin film properties, and to establish a comprehensive system for assessing the reliability of multi-layered semiconductor packaging structures. Several responsibilities are summariezd:* *-Development of mechanical property measurement equipment of nanolayer* *-Test system development of the interfacial adhesion in multi-nano layer* *-Development of advanced multiscale, multiphysical model connecting microscale and macroscopic deformation modes and dominant failure* *-Developing machine learning approaches based on experimental and rheoretical results, enabling more accurate predictions of device reliability and performance* *-Solution development by combining experimental results and nano mechanics model*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The project will attain potentially transformative new knowledge on the deformation mechanism of plasticity and fracture, enabled by the recent advances in multi-scale computational modeling and experimental techniques, in-situ microscopy including multiphysical analysis and machine learning approach. In particular, integrated multiscale modeling will clarify how microstructural features—such as dislocation activity, GB/PL-effects, and void evolution—govern damage initiation and failure pathways, thereby improving mechanistic understanding of microstructure-driven failure in multilayer interconnectors. An experimentally validated predictable model based on machine learning approach will be established and will enable more accurate predictions of device reliability and performance. Advanced chiplet technology demands greater heterogeneity in materials, interfaces, and length scales, while product cycles are increasingly compressed—making accurate, physics-based reliability prediction even more critical. Therefore, a comprehensive understanding of mechanical properties of relevant materials and failure mechanisms in the metallization layer in small scale provides invaluable information, that can help enginneers to identify potential failure points, understand performance degradation, and predict the lifespan of microchips in another level. Combined with theoretical and experimental approaches, it provides a robust framework for reliability engineering in multilayered integrated circuits and this can minimize reliability risk at the design stage. It is expected to have great impact in cutting-edge wafe fabrication and advanced packaging development. In addition, the developed nano-scale stress measurement system, enhanced by the nano-mechanics models and advanced experimental results can be used as next-generation realiability evaluation tool for multi-layer interconnectors from semiconductor microchips.*

6. References

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization